CBSE Class 12 Physics — Half-Yearly Predicted Paper (Set 2 of 10)

Section A: 1-Mark Questions (MCQ/Assertion-Reason)

Q1. A point charge q is placed at the exact centre of a closed cube. The total electric flux passing through one single face of the cube will be: (a) q/ϵ_0 (b) $q/6\epsilon_0$ (c) $q/4\epsilon_0$ (d) Zero [PYQ 2019] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q2. A positively charged hollow metal sphere of radius R is in electrostatic equilibrium. The electric potential at its centre is: (a) Zero (b) Less than the potential at its surface (c) Same as the potential at its surface (d) Greater than the potential at its surface [PYQ 2025-26 SQP] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q3. Kirchhoff's loop rule (voltage rule) is a direct consequence of the law of conservation of: (a) Charge (b) Energy (c) Momentum (d) Angular momentum [PYQ 2021] | **[Chapter 3: Current Electricity | Confidence: Very High]**

Q4. A charged particle is projected along the axis of a current-carrying circular loop. Which of the following statements is physically true? (a) The acceleration of the charged particle depends on the velocity with which it is projected. (b) The charged particle will accelerate radially outwards. (c) The charged particle will move with a constant velocity. (d) The charged particle will trace a helical path. [PYQ 2025-26 SQP] | **[Chapter 4: Moving Charges & Magnetism | Confidence: High]**

Q5. Two straight wires of the exact same length are shaped into a square of side a and a circle of radius r. If they carry the exact same current I, the ratio of their respective magnetic dipole moments (M_{square}/M_{circle}) is: (a) $2:\pi$ (b) $\pi:2$ (c) $\pi:4$ (d) $4:\pi$ [PYQ 2022] | **[Chapter 5: Magnetism & Matter | Confidence: High]**

Q6. Two small identical magnets are allowed to fall freely from rest. One falls through a long vertical copper pipe, and the other falls freely in air through the exact same vertical distance. The time taken to reach the ground will be: (a) Same in both the cases. (b) More for the magnet falling in air. (c) More for the magnet falling through the copper pipe. (d) Cannot be determined without mass values. [PYQ 2025-26 SQP] | **[Chapter 6: Electromagnetic Induction | Confidence: Very High]**

Q7. In a series LCR circuit, the phase difference between the applied AC voltage and the current at electrical resonance is: (a) $\pi/2$ (b) $\pi/4$ (c) π (d) Zero [Practice] | **[Chapter 7: Alternating Current | Confidence: Very High]**

Q8. Which of the following electromagnetic waves is predominantly used in radar systems for aircraft navigation? (a) Infrared rays (b) Microwaves (c) Ultraviolet rays (d) Radio waves (long-wavelength) [PYQ 2016, 2020] | **[Chapter 8: Electromagnetic Waves | Confidence: Very High]**

Q9. A symmetric biconvex lens of glass (refractive index 1.5) has a focal length f in air. If it is completely immersed in water (refractive index 1.33), its new focal length will: (a) Decrease (b) Remain unchanged (c) Increase (d) Become negative [PYQ 2020] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Q10. In a single slit diffraction experiment, if the width of the slit is strictly halved, the angular width of the central maximum will: (a) Become half (b) Remain unchanged (c) Become double (d) Become four times [Practice] | **[Chapter 10: Wave Optics | Confidence: High]**

Q11. The correct relationship between the linear momentum p of a photon and its wavelength λ is: (a) $p \propto \lambda$ (b) $p \propto 1/\lambda$ (c) $p \propto \lambda^2$ (d) $p \propto 1/\lambda^2$ [PYQ 2020] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Q12. Einstein's photoelectric equation, $K_{max} = h\nu - \Phi$, is a direct mathematical statement of the law of conservation of: (a) Linear momentum (b) Angular momentum (c) Charge (d) Energy [PYQ 2017] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

For Questions 13 to 16, two statements are given — one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c), and (d) as given below. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false and R is also false.

Q13. Assertion (A): Two electric field lines never intersect each other. **Reason (R):** If they intersected, there would be two distinct directions of the electric field at the point of intersection, which is physically impossible. [PYQ 2015] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q14. Assertion (A): The macroscopic electrical resistance of a uniform wire strictly increases when its length is doubled by stretching it. **Reason (R):** Stretching a wire doubles its length and halves its cross-sectional area, increasing its resistance by a factor of 4. [Practice] | **[Chapter 3: Current Electricity | Confidence: High]**

Q15. Assertion (A): The critical angle of light passing from glass to air is minimum for violet colour. **Reason (R):** The refractive index of glass is maximum for violet light compared to other visible colours. [PYQ 2025-26 SQP] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Q16. Assertion (A): Two independent light sources emitting light waves of identical wavelength can easily act as coherent sources. **Reason (R):** Coherent sources must strictly emit waves that maintain a zero or constant phase difference over time. [PYQ 2025-26 SQP] | **[Chapter 10: Wave Optics | Confidence: Very High]**

Section B: 2-Mark Questions (Very Short Answer)

Q17. An electric dipole of dipole moment $\vec{p}$ is placed in a uniform electric field $\vec{E}$. Write the expression for the maximum torque experienced by the dipole. Also, determine the work done in turning the dipole from its stable equilibrium position to its unstable equilibrium position. [PYQ 2017] | **[Chapter 1: Electric Charges & Fields | Confidence: High]**

Q18. A battery of unknown emf E and internal resistance $3\ \Omega$ is connected to an external resistor of $17\ \Omega$. If the circuit current is measured to be $0.5\ \mathrm{A}$, calculate the exact value of the emf of the battery and the terminal voltage across the external resistor. [Practice] | **[Chapter 3: Current Electricity | Confidence: High]**

Q19. A pair of adjacent coils has a mutual inductance of $1.5\ \mathrm{H}$. If the steady current in one coil changes uniformly from 0 to $20\ \mathrm{A}$ in exactly $0.5\ \mathrm{s}$, what is the total change of magnetic flux linkage with the other secondary coil? [PYQ 2016, 2022] | **[Chapter 6: Electromagnetic Induction | Confidence: Very High]**

Q20. State any two important, practical uses of (i) Infrared waves, and (ii) Ultraviolet rays. [PYQ 2014, 2018] | **[Chapter 8: Electromagnetic Waves | Confidence: Very High]**

Q21. Define the term "stopping potential" in the context of the photoelectric effect. Does the stopping potential depend on the intensity of the incident light? Give a brief reason. [PYQ 2019] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Section C: 3-Mark Questions (Short Answer)

Q22. Deduce the mathematical expression for the electrostatic potential energy of a system consisting of two point charges q_1 and q_2 located at position vectors $\vec{r}_1$ and $\vec{r}_2$ respectively, placed in an external uniform electric field $\vec{E}$. [PYQ 2020] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q23. State Kirchhoff's two rules for electrical circuits. Using these rules, derive the standard balance condition for a Wheatstone bridge ($\frac{P}{Q} = \frac{R}{S}$). [PYQ 2015] | **[Chapter 3: Current Electricity | Confidence: Very High]**

Q24. A moving coil galvanometer of resistance G gives full-scale deflection for a current I_g. Explain mathematically how it can be converted into: (i) an ammeter of range 0 to I (where $I > I_g$). (ii) a voltmeter of range 0 to V. [PYQ 2014, 2020] | **[Chapter 4: Moving Charges & Magnetism | Confidence: High]**

Q25. State Bohr's quantization condition for angular momentum. Using this postulate, deduce the expression for the magnetic dipole moment of an electron revolving in the n^{th} stable orbit of a hydrogen atom. [PYQ 2015] | **[Chapter 5: Magnetism & Matter | Confidence: Medium]** *(Note: Bohr's model is formally in 'Atoms', but magnetic moment of revolving electron is covered under Magnetism Chapter 4/5. Keeping as standard magnetic moment derivation).*

Q26. Draw a neatly labelled schematic diagram of a step-up transformer. Explain its fundamental working principle. State any two distinct causes of energy loss in practical transformers. [PYQ 2016, 2014] | **[Chapter 7: Alternating Current | Confidence: Very High]**

Q27. Two thin convex lenses of focal lengths f_1 and f_2 are placed coaxially in direct physical contact with each other. Obtain the rigorous mathematical expression for the equivalent focal length of this combination. Define the equivalent optical power. [PYQ 2020] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Q28. State Huygens' principle of secondary wavelets. Using Huygens' geometrical construction, deduce the laws of reflection of light from a plane reflecting surface. [PYQ 2019, 2023] | **[Chapter 10: Wave Optics | Confidence: Very High]**

Section D: 4-Mark Questions (Case-Study Based)

Q29. Case Study: Temperature Dependence of Electrical Resistivity The electrical resistance of a conductor is determined by its physical dimensions and the inherent resistivity of the material. Resistivity, in turn, depends heavily on the microscopic relaxation time of charge carriers. When the temperature of a metallic conductor is raised, the thermal agitation of its atoms increases. The atoms vibrate with greater amplitude, causing the free electrons to collide more frequently. Consequently, the average relaxation time (τ) strictly decreases, leading to an increase in resistivity. However, in semiconductors, an increase in temperature breaks covalent bonds, exponentially increasing the number density of charge carriers (n), which drastically lowers their resistivity. The fractional change in resistivity per degree rise in temperature is termed the temperature coefficient of resistivity (α). (A) How does the drift velocity of free electrons in a metallic wire change if its temperature is increased while keeping the applied voltage constant? (1 Mark) (B) Give a scientific reason why the temperature coefficient of resistivity (α) is negative for semiconductors like Silicon and Germanium. (1 Mark) (C) A silver wire has an electrical resistance of $2.1\ \Omega$ at $27.5^\circ\mathrm{C}$ and a resistance of $2.7\ \Omega$ at $100^\circ\mathrm{C}$. Determine the exact temperature coefficient of resistivity (α) of silver. (2 Marks) [Practice / PYQ 2019] | **[Chapter 3: Current Electricity | Confidence: Very High]**

Q30. Case Study: The Compound Microscope A compound microscope is a classic optical instrument utilized for obtaining highly magnified images of extremely tiny objects. It consists essentially of two converging lenses: an objective lens of very short focal length and small aperture facing the object, and an eyepiece of slightly larger focal length and aperture facing the eye. The objective forms a real, inverted, and magnified image of the object. This intermediate image acts as a real object for the eyepiece, which functions like a simple magnifier to form the final, highly magnified virtual image. (A) Why is the focal length of the objective lens deliberately kept very short in a compound microscope? (1 Mark) (B) Write the formula for the total magnifying power of a compound microscope when the final image is formed precisely at the least distance of distinct vision (D). (1 Mark) (C) A compound microscope utilizes an objective lens of focal length $1.0\ \mathrm{cm}$ and an eyepiece of focal length $5.0\ \mathrm{cm}$. The distance between the two lenses is physically set to $15.0\ \mathrm{cm}$. If the final image is formed at infinity (normal adjustment), calculate the required distance of the object from the objective lens. (2 Marks) [PYQ 2015, 2020] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Section E: 5-Mark Questions (Long Answer)

Q31. (A) State Gauss's theorem in electrostatics. (B) Apply Gauss's theorem to derive the mathematical expression for the electric field intensity due to a uniformly charged thin spherical shell of radius R and total charge Q, at a point: (i) strictly outside the shell ($r > R$). (ii) strictly inside the shell ($r < R$). (C) Plot a properly labelled graph showing the variation of the electric field intensity E as a function of the distance r from the geometric centre of the spherical shell. [PYQ 2015, 2018] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q32. (A) Draw a fully labelled phasor diagram for a series LCR circuit connected to an AC source of voltage $V = V_0 \sin(\omega t)$. (B) Derive the mathematical expression for the average power dissipated over one complete cycle in this series LCR circuit. (C) Define the 'Power Factor' of an AC circuit. State its exact numerical value for: (i) a purely resistive circuit, and (ii) a purely inductive circuit. (D) A series LCR circuit with $L = 2.0\ \mathrm{H}$, $C = 32\ \mu\mathrm{F}$, and $R = 10\ \Omega$ is connected to a variable frequency $200\ \mathrm{V}$ AC supply. Calculate the precise resonant angular frequency (ω_r) of the circuit. [PYQ 2014, 2019] | **[Chapter 7: Alternating Current | Confidence: Very High]**

Q33. (A) Describe Young's double-slit experiment layout to produce an interference pattern on a screen due to a monochromatic source of light. (B) Deduce the mathematical expression for the path difference between two interfering light waves, and hence derive the formula for the fringe width (β). (C) State any two distinct differences between the interference pattern obtained in Young's double-slit experiment and the diffraction pattern obtained due to a single narrow slit. (D) If the entire experimental apparatus of Young's double-slit experiment is completely immersed in water (refractive index $4/3$), what exact effect will this have on the fringe width? Justify your answer. [PYQ 2016, 2019] | **[Chapter 10: Wave Optics | Confidence: Very High]**

Answer Key

Q1: (b) $q/6\epsilon_0$. Total flux is q/ϵ_0 by Gauss's Law. Symmetrically distributed across 6 identical faces, flux per face is $q/6\epsilon_0$. **Q2:** (c) Same as the potential at its surface. The electric field inside a charged conductor is zero, so potential is constant throughout and equals the surface potential. **Q3:** (b) Energy. The net potential drop around a closed loop is zero, representing energy conservation for charges moving in the circuit. **Q4:** (c) The charged particle will move with a constant velocity. The magnetic field on the axis of a circular loop is parallel to the axis. Thus, velocity $\vec{v}$ is parallel to $\vec{B}$. Force $\vec{F} = q(\vec{v} \times \vec{B}) = 0$. Zero force implies zero acceleration (constant velocity). **Q5:** (c) $\pi : 4$. Area of square $A_s = (L/4)^2 = L^2/16$. Area of circle $A_c = \pi(L/2\pi)^2 = L^2/4\pi$. Ratio of moments $M_s/M_c = (I \cdot A_s)/(I \cdot A_c) = (L^2/16)/(L^2/4\pi) = 4\pi/16 = \pi/4$. **Q6:** (c) More for the magnet falling through the copper pipe. Falling through copper induces eddy currents. By Lenz's law, the magnetic field of eddy currents opposes the relative motion, reducing downward acceleration ($a < g$). **Q7:** (d) Zero. At resonance, inductive reactance equals capacitive reactance ($X_L = X_C$). The circuit behaves purely resistively, so voltage and current are in exactly the same phase. **Q8:** (b) Microwaves. They are highly directional and suitable for radar systems. **Q9:** (c) Increase. The relative refractive index of glass with respect to water ($1.5/1.33 = 1.13$) is less than that with respect to air ($1.5/1 = 1.5$). By Lens Maker's formula, a smaller relative index means a larger focal length. **Q10:** (c) Become double. Angular width of central maximum $\theta = 2\lambda/a$. If slit width a is halved, angular width strictly doubles. **Q11:** (b) $p \propto 1/\lambda$. By de Broglie relation, $p = h/\lambda$. **Q12:** (d) Energy. The equation equates incident photon energy to the work function plus the maximum kinetic energy, balancing total energy.

Q13: (a) Both A and R are true and R is the correct explanation of A. **Q14:** (a) Both A and R are true and R is the correct explanation of A. (Volume $V = A \cdot L$ is constant; $L \to 2L$ means $A \to A/2$. $R \propto L/A \to 4R$). **Q15:** (a) Both A and R are true and R is the correct explanation of A. ($\sin C = 1/\mu$. Violet light has the maximum refractive index μ, therefore its critical angle C is strictly minimum). **Q16:** (d) A is false and R is true. Independent sources undergo random, rapid phase shifts and can *never* be coherent. Coherent sources must have a constant phase difference (R is true).

Q17: Maximum torque $\tau_{max} = pE\sin(90^\circ) = pE$. Work done $W = \int_{0^\circ}^{180^\circ} pE\sin\theta d\theta = pE[-\cos\theta]_0^{180^\circ} = pE[-(-1) - (-1)] = 2pE$. **Q18:** Total resistance = $R + r = 17 + 3 = 20\ \Omega$. Emf $E = I(R + r) = 0.5 \times 20 = 10\ \mathrm{V}$. Terminal voltage $V = IR = 0.5 \times 17 = 8.5\ \mathrm{V}$. **Q19:** Change in current $\Delta I = 20 - 0 = 20\ \mathrm{A}$. Change in flux linkage $\Delta\Phi = M\Delta I = 1.5 \times 20 = 30\ \mathrm{Wb}$. **Q20:** (i) Infrared: Used in night vision goggles, TV remotes, and treating muscular strains. (ii) Ultraviolet: Used in water purifiers to kill germs, and in LASIK eye surgery. **Q21:** Stopping potential is the minimum negative (retarding) potential applied to the anode at which the photoelectric current becomes strictly zero. No, it does *not* depend on the intensity of incident light; it depends only on the frequency of incident light and the nature of the material.

Q22: The work done to bring q_1 from infinity to $\vec{r}_1$ is $W_1 = q_1 V(\vec{r}_1)$. The work done to bring q_2 to $\vec{r}_2$ involves work against the external field ($q_2 V(\vec{r}_2)$) and work against the field of q_1 ($\frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r_{12}}$). Total potential energy $U = W_1 + W_2 = q_1 V(\vec{r}_1) + q_2 V(\vec{r}_2) + \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r_{12}}$. **Q23:** Rules: (1) Junction Rule ($\Sigma I = 0$): Based on charge conservation.

(2) Loop Rule ($\Sigma \Delta V = 0$): Based on energy conservation. Bridge derivation: Draw a Wheatstone bridge with arms P, Q, R, S and Galvanometer G. For balance condition, $I_g = 0$. Apply Loop rule to the two closed loops to get $I_1 P = I_2 R$ and $I_1 Q = I_2 S$. Dividing them gives $P/Q = R/S$. **Q24:** (i) Ammeter: A small shunt resistance S is connected strictly in *parallel* to the galvanometer. $S = \frac{I_g \times G}{I - I_g}$. (ii) Voltmeter: A very high resistance R is connected strictly in *series*. $R = \frac{V}{I_g} - G$. **Q25:** Bohr's condition: Angular momentum $L = mvr = nh/2\pi$. Magnetic moment $M = I \times A = (e/T) \times (\pi r^2)$. Since $T = 2\pi r/v$, $M = (ev/2\pi r) \times (\pi r^2) = evr/2$. Multiply and divide by m: $M = e(mvr)/2m$. Substitute L: $M = e(nh/2\pi)/2m = neh/4\pi m$. **Q26:** Diagram showing primary coil (fewer turns) and secondary coil (more turns) on a laminated iron core. Principle: Mutual induction (changing AC in primary induces emf in secondary). Energy losses: (1) Copper loss (Joule heating $I^2 R$ in wires), (2) Iron loss (Eddy currents in core), (3) Flux leakage, (4) Hysteresis loss. **Q27:** For the first lens, image is formed at v'. $\frac{1}{v'} - \frac{1}{u} = \frac{1}{f_1}$. This acts as a virtual object for the second lens: $\frac{1}{v} - \frac{1}{v'} = \frac{1}{f_2}$. Adding the equations: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2}$. For the equivalent lens: $\frac{1}{v} - \frac{1}{u} = \frac{1}{F}$. Hence, $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$. Optical Power $P = 1/F = P_1 + P_2$. **Q28:** State Huygens' principle. Draw plane wavefront AB incident at angle i on a reflecting surface. By the time point B reaches C (distance vt), secondary wavelet from A spreads to radius vt. Draw tangent CD from C to this wavelet. Prove congruent triangles $\triangle ABC \cong \triangle ADC$, thereby proving $\angle i = \angle r$.

Q29: (A) Decreases. (τ drops due to increased collisions; $v_d = eE\tau/m$, so v_d decreases). (B) In semiconductors, increasing temperature dramatically breaks bonds, releasing exponentially more free charge carriers (n), dropping resistance heavily. Thus, resistance decreases as temperature rises, making α strictly negative. (C) $R_T = R_0[1 + \alpha(T - T_0)] \Rightarrow 2.7 = 2.1[1 + \alpha(100 - 27.5)] \Rightarrow 2.7/2.1 = 1 + 72.5\alpha \Rightarrow 1.285 = 1 + 72.5\alpha \Rightarrow \alpha = 0.285/72.5$. **Q30:** (A) To achieve high initial magnification ($m_o \approx L/f_o$), the objective focal length must be small. (B) $M = m_o \times m_e = \left(\frac{L}{f_o}\right) \times \left(1 + \frac{D}{f_e}\right)$. (C) Final image at infinity means intermediate image forms exactly at the focus of the eyepiece ($v_e = f_e = 5\ \mathrm{cm}$). Thus, $v_o = L_{tube} - f_e = 15 - 5 = 10\ \mathrm{cm}$. Using lens formula for objective: $1/v_o - 1/u_o = 1/f_o \Rightarrow 1/10 - 1/u_o = 1/1 \Rightarrow 1/u_o = 1/10 - 1 = -9/10 \Rightarrow u_o = -10/9 = -1.11\ \mathrm{cm}$.

Q31: (A) Total electric flux through a closed Gaussian surface is $1/\epsilon_0$ times the net charge enclosed. (B) (i) For $r > R$, consider spherical Gaussian surface of radius r. $\oint E \cdot dS = E(4\pi r^2) = Q/\epsilon_0 \Rightarrow E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$. (ii) For $r < R$, Gaussian surface encloses strictly zero charge. $E(4\pi r^2) = 0 \Rightarrow E = 0$. (C) Graph: $E = 0$ from $r = 0$ to $r = R$. At $r = R$, E jumps to max value $\frac{1}{4\pi\epsilon_0}\frac{Q}{R^2}$, then strictly decays as $1/r^2$ for $r > R$. **Q32:** (A) Draw a voltage-current phasor diagram with V_R in phase with I, V_L leading by 90°, and V_C lagging by 90°. (B) Instantaneous power $P = VI = V_0 \sin(\omega t) \times I_0 \sin(\omega t - \phi)$. Integrating over one cycle and dividing by T yields $P_{avg} = V_{rms} I_{rms} \cos\phi$. (C) Power Factor = $\cos\phi = R/Z$. (i) Purely resistive: $\phi = 0 \Rightarrow \cos\phi = 1$. (ii) Purely inductive: $\phi = 90^\circ \Rightarrow \cos\phi = 0$. (D) $\omega_r = 1/\sqrt{LC} = 1/\sqrt{2 \times 32 \times 10^{-6}} = 1/\sqrt{64 \times 10^{-6}} = 1/(8 \times 10^{-3}) = 1000/8 = 125\ \mathrm{rad/s}$. **Q33:** (A) Describe the two coherent slits S_1 and S_2 illuminated by a single source, forming bands on a screen at distance D. (B) Path difference $\Delta x = S_2 P - S_1 P \approx yd/D$. For bright fringes, $yd/D = n\lambda \Rightarrow y_n = n\lambda D/d$. Fringe width $\beta = y_{n+1} - y_n = \lambda D/d$. (C) (1) Interference has fringes of equal width; diffraction central maximum is double the width of others. (2) Interference bright fringes have uniform maximum intensity; diffraction maxima strictly drop in intensity. (D) Immersing in water reduces the wavelength to $\lambda' = \lambda/\mu = \lambda/(4/3) = 3\lambda/4$. Since $\beta \propto \lambda$, the new fringe width decreases to $3/4$ of its original value.

CBSE Class 12 (2027) — Half-Yearly Predicted Paper (80% syllabus). AI-generated via PYQ/Blueprint analysis, not actual exam questions.